

How Artificial Intelligence Shapes Social Change: Adoption, Power, and Governance

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1 Abstract

Public and academic debates about Artificial Intelligence (AI) shift between polarized expectations. Optimistic accounts anticipate far-reaching societal transformation, while skeptical perspectives dismiss AI as technological hype with limited real-world consequences beyond marginal efficiency gains and new inequalities. This article argues that both positions obscure how technologies actually shape societies. Drawing on insights from innovation studies, organizational sociology, and theories of social power, it advances a general theory of AI-induced social change that conceptualizes impact as contingent, stepwise, and mediated. The theory models AI's societal effects as a causal chain linking scientific breakthroughs, technological innovation, organizational and individual adoption, uneven gains in relative advantage, shifts in power within and between organizational networks, governance responses, and structural constraints. Central to this framework is the claim that AI alters social outcomes primarily by enabling some actors to pursue their goals more effectively than others, thereby generating localized power shifts. Whether such shifts translate into broader social change depends on subsequent processes, including regulatory interventions, political acceptance, and material, institutional, and societal constraints. By integrating micro-, meso-, and macro-level processes, the framework helps explain why AI can produce significant change in specific domains without leading to immediate or uniform transformation at the societal level. It also clarifies why the absence of macro-level change does not imply the absence of meaningful impact. The theory provides a framework to support cumulative research on AI's societal consequences and to inform more precise and grounded public discussions of AI-enabled social change.

Keywords: Artificial Intelligence; Power shifts; Technology adoption; Governance

2 What to expect when expecting AI-enabled social change?

The public launch of Open AI's language model interface *Chat GPT* raised public awareness of possible uses of generative models and their application to various tasks and fields. The subsequent development of new models, applications, and services reinforced early enthusiasm and fears. Further, as experiences with AI and evidence of wide applications mounted, so did expectations of AI-enabled social change. These expectations include accounts of deep societal transformation for both the positive and the negative (Amodei, 2024; Kokotajlo et al., 2025; Lee & Quifan, 2021; Suleyman & Bhaskar, 2023). But looking closely, technologically-enabled social change is neither automatic, nor obvious.

New technologies can lead to social change by allowing people to do new things, or to do old things better, quicker, or cheaper. For example, in warfare new technologies and their applications have allowed for shifts in competitive advantage allowing early movers to gain on or replace incumbents initiating sometimes severe power shifts and associated social change (McNeill, 1982). In politics, new technologies have allowed new actors to emerge capitalizing for example on advances in information technologies and challenge established actors and organizations (Bimber, 2003; Headrick, 1992). Economics shows that countries aiming to realize opportunities by new technologies have been shown to broadly adjust institutions to provide a better foundation for the use of new technologies in society and to have them adopted widely (Ding, 2024).

Specific technological inventions can also contribute to social change. The development of agriculture incentivized people to settle and in turn made them more visible and better taxable by central bureaucracies, leading in turn to the emergence of strong central states with resources to spend beyond mere subsistence tasks in turn leading to cultural and civilizational achievements (Scott, 2017). The invention of the printing press and associated institutional adaptations led to a stabilization of established knowledge, supporting the establishment of cumulative knowledge. By lowering the costs of information production, it also contributed to the successful challenge of authority by allowing challengers of the status quo voice and information distribution (Eisenstein, 1979; Kaufmann, 2019). The invention and broad availability of the birth control pill in the U.S. from the late 1960s onwards contributed to women marrying later in life, choosing more frequently to pursue professional degrees in higher education (Goldin & Katz, 2002), and to an increased presence of women in the labor force (Bailey, 2006). This in turn clearly changed the role of women in society and led to subsequent shifts in personal lives, economics, culture, and politics.

These examples illustrate that some new technologies have led to social change. But they did so contingently and in close interaction with existing practices, structures, and power relationships. To better understand the impact of AI in society, we can model this process according to patterns observed with prior technologies.

Expectations for AI induced social change are strong. By lowering the costs of inferring missing information, AI could contribute to new business models and greater opportunities for economic foresight (Agrawal et al., 2018/2022). New opportunities for the automation of tasks in creative and knowledge work might increase productivity (Brynjolfsson et al., 2025) but over time might also lead to the replacement of well paid knowledge workers, requiring new sources and mechanisms for the redistribution of AI-enabled economic gains (Acemoglu & Johnson, 2023). AI-enabled foresight and robotics are expected to change warfare, raising fears that winning or losing the AI

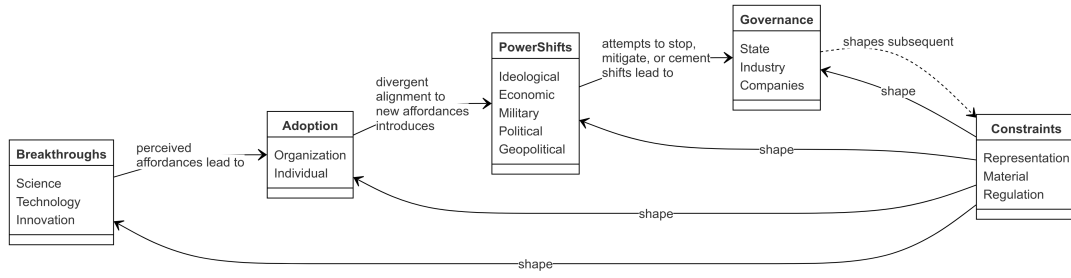


Figure 1: The stepwise model of AI-driven social change.

race might lead to a significant rebalancing of military power between countries (Brose, 2020; Buchanan & Imbrie, 2022; Scharre, 2023). For democracies, losing the AI race to autocracies can thus be expected to have strong consequences for their capabilities to project power or defend their interests. Within democracies, AI is also expected to lead to changes within politics, political competition, and governance. The automation of tasks impacts the work of political organizations potentially empowering resource-poor but adaptive organizations that more often than not are challengers of the political status quo (Jungherr et al., 2026). New AI-enabled sources of insight, foresight, and service delivery can change practices and expectations toward governance (Farrell, 2025; Schneier & Sanders, 2025).

This is just a small selection of examples for different societal fields where AI is expected to lead to significant change, for better or worse. This article presents a theoretical framework for the analysis and prediction of AI-enabled social change. It presents an original model linking the emergence of technological inventions to social change and institutional response. This provides both an analytical framework for the systematic, stepwise analysis of AI’s impact on society in various fields and the prediction of impacts. This offers new opportunities for cumulative social science work on AI’s impact across fields and developmental steps, helping in turn societies to better foresee and shape AI development and impact.

3 The model: From technological breakthroughs to social change

To understand, explain, and shape the impact of AI on society, we need to start with a model. The dynamics of AI development, adoption, and regulation resemble that of other technologies and can be modeled as a series of connected steps. This process connects different societal actors, fields, and analytical levels.

The impact of AI on society is a dynamic process with many preceding events and factors. But for sake of illustration, we can start the discussion of the model of AI-driven social change with the innovation stage.

In the context of AI, we need to consider two connected but distinct elements of *innovation*. The scientific breakthroughs that lie at the foundation of the current wave of generative AI and the translation of these breakthroughs into new tools, products, and processes that allow different actors to capitalize on new opportunities provided by AI. Both the scientific as well as the translation step are necessary for innovation to happen.

But once these processes are started, we see innovations to create new affordances for

actors willing to adopt them.

Just because technological innovations are available does not necessarily mean, they are picked up. To explain *adoption* decisions by both organizations and individuals, we need to look at the structural and psychological processes that lead actors to recognize innovations, perceive affordances for them to pursue their goals better, and to include them in their practices, processes, and products.

Successful adoption of AI and AI-enabled systems will be easier for some actors than others. Either through willingness for early adoption, experimentation, and ability to adjust existing structures some actors will be more ready to capitalize on AI-driven affordances than others. This means that they are relatively more advantaged by AI than others. If this relative advantage is great it might even lead to *power shifts* within societal fields.

There are various different areas of social power within and between societies. Following Michael Mann's theory of social power (Mann, 1986/2012a, 1993/2012b, 2012c, 2013), we can distinguish four sources of social power: ideological, economic, military, and political. This allows us to conceptualize and map potential AI-driven power-shifts within society. To account for potential power shifts between societies, we also should include geopolitics. Within each of these fields, shifts in the relative influence of specific actors due to their relatively better alignment with affordances provided by AI might lead to shifts in relative power between actors in these areas or the relative importance these sources of social power have within societies at any given point in time.

AI-driven affordances and divergent relative alignments to capitalize on them might therefore lead to substantive shifts of who holds power within and between societies. These shifts, actual or expected, create demands for *governance*. Depending on the relative influence of actors benefiting or finding themselves challenged by AI, demands for governance interventions by states, industry, or companies shift. Those threatened by AI are likely to ask for restrictive interventions slowing down or hindering the development and use of AI. Conversely, those benefiting from AI demand the opposite of regulators.

The resulting regulatory environment will mediate the impact of AI on society and in turn shape the conditions for any future scientific breakthroughs and connected innovations. The model accounts for the circular nature of the underlying process. Scientific breakthroughs are translated into innovations that some actors manage to capitalize on. Associated shifts in relative power – perceived or real – lead to changes in governance rules, which then again provide the basis for future breakthroughs and a new iteration of the circle.

But governance is not the only *constraint* within this circle. In fact, every step happens in the context of various pre-existing constraints. As discussed, this includes regulation. But there are other constraints we need to consider as well. This includes the degree to which relevant aspects of the world are represented or representable in data but also material constraints that limit or enable the use of AI in a given space, such as available computing infrastructures, chips, energy, or pertinent human talent.

This simple model provides a promising framework by which we can map and model the impact of AI on society. In the following sections I will explicate the model and its steps in greater detail.

4 Breakthroughs: Science, technology, and innovation

The first step in explaining the impact of AI on society lies in explaining the conditions under which AI-related scientific breakthroughs and technological innovations are achieved. Science, technology, and innovation are deeply intertwined but distinct phenomena. Treating them as distinct phenomena allows us to tease apart their determinants, effects, and contextual constraints.

Science is the process by which we learn about the world. This includes establishing facts about the world, developing generalizations about sets of facts and phenomena, and providing explanations for them (Simon, 2002).

Technology is a way for people or organizations to achieve a purpose. Technologies can be devices, methods, or processes. The associated purposes can be explicitly designed and static, or they can be emergent and dynamic. For example, computers started out having a specific narrowly defined purpose as automated calculating machines. Over time this purpose has been broadened as they found applications in the pursuit of countless other goals (Arthur, 2009, p. 19f.).

Science and technology are clearly connected. New knowledge about the world allows for the development of new means to pursue existing goals or even establish new ones (Bush, 1945). But improving technology and their contribution in the pursuit of goals can also go beyond the current state of scientific understanding of the world. In these cases practice precedes understanding, pointing out areas where scientific knowledge has to follow technological capabilities (Stokes, 1997). An obvious example for this technology-first science-second relationship is the use of deep learning in AI (LeCun et al., 2015). Deep learning techniques underly many applications that help developers pursue specific goals, such as image recognition, content generation, or strategic game play. Deep learning is a technology that allows the pursuit of goals. At the same time, scientific understanding of the exact process by which these techniques achieve these purposes is lacking. The relationship between science and technology is therefore circular and developments in either area can precede those in the other.

Innovation is the process by which established processes to achieve specific outcomes are reconfigured to account for novel knowledge about the world or the availability of novel technology (Schumpeter, 1911/1997, p. 99–110). Innovation can take different forms including new production processes, substituting elements of the production chain with improved, or the reorganization of workflows (Kline & Rosenberg, 1986, p. 279). Originally, the term has been used in the context of production functions. But we can transpose the concept to other areas (Grove, 1983). Collective and individual actors pursue goals, in other words, they follow processes, practices, or actions to create outputs. For a company a desired output can be a product to sell, for a university a degree to hand out, for a politician a vote won, etc. To achieve these goals actors explicitly – or more often implicitly – follow output functions. Accordingly, AI-enabled innovation happens across fields by allowing actors to create new output functions, reconfigure them, or exchange elements in their existing process of output generation.

We have discussed deep learning as a technology allowing us to use neural networks in the pursuit of goals, such as image recognition. Including deep learning technology for image recognition within a service or process represents an innovation. For example, using deep learning in a photo app dedicated to labeling images is a commercial innovation. Using deep learning for image recognition to label facial expression in images of people for research is a scientific innovation. AI-driven innovation can be small – as in

the replacement of manual process through automated techniques – or large – for example in the creation of new products and services made possible through the creative use of cheap automation (Agrawal et al., 2018/2022).

Science, technology, and innovation offer us three points of potential breakthroughs where AI can start to influence society. Understanding the conditions and constraints for scientific research into AI allows us to increase our understanding of its workings and limitations. Documenting and explaining the instrumental use of AI in novel technologies to achieve goals, allows us to identify the conditions for its successful implementation. While the close look at the uses of AI and AI-enabled technologies in established production chains or processes, will allow us to model its impact better and in greater resolution.

This conceptual parsimony allows us to address different aspects of how AI is introduced in societies. For example, there are rich traditions in the work on the interconnection between scientific research, technological progress, and innovation (Edquist, 1997; Narayanamurti & Tsao, 2021; Rosenberg, 1983, 1994). These works point out that these processes follow different logics and determinants, but are closely – if not linearly – connected. We should build on these traditions with a special focus on the specifics of AI-related research, technology, and innovation.

At its core the science of current wave of AI is math (Ananthaswamy, 2024). Algorithms identify recurring patterns (relationships between variables), translate them into functions, and use those to predict variables when they are unknown, such as likely outcomes of future actions. These functions can be comparatively simple, as in the case of regression analyses, or they can be complex, when represented through neural networks. Neural networks are at the heart of deep learning and fuel the current wave of generative AI. Their mathematical foundations go back to the 1950s (Nilsson, 2010, p. 62–80). But it took until the mid 2000s for the potential of multi-layered neural networks and deep learning to become obvious, when researchers enabled by access to greater computational power, unprecedented amounts of data, and new approaches to the architecture of multilayered neural networks solved a set of core problems with the approach (LeCun et al., 2015; Schmidhuber, 2015). These insights were implemented in new technologies, designed to solve specific tasks, such as speech recognition (Hinton et al., 2012), natural language processing (Goldberg, 2016), and computer vision (Lakshmanan et al., 2021). Neural networks are also at the core of the current wave of generative AI in the form of transformer models (Vaswani et al., 2017). These models power technologies that allow the production of text, image, audio, and visual content (Brown et al., 2020; Parmar et al., 2018; Ramesh et al., 2022). More recently, these capabilities have been extended through the development of agentic AI systems, which combine generative models with planning, memory, and tool-use components to autonomously pursue goals over extended sequences of actions. Rather than merely producing outputs in response to prompts, such systems are designed to decompose tasks, evaluate intermediate outcomes, and adapt strategies based on feedback, blurring the line between prediction and action (Schick et al., 2023; Yao et al., 2023). While current agentic systems remain constrained and brittle, their emergence raises important questions about delegation, control, and responsibility when AI systems are embedded more deeply in organizational and social processes (Catalini et al., 2026; Renieris et al., 2025).

These approaches are currently being integrated in work processes in various fields, such as the creative industry, medicine, coding, or education. These attempts at integrating AI in existing production processes and practices are innovations. Although, it

has to be said, the degree to which this integration provides improvement is very much contested and open to investigation.

Using the concepts introduced above, we can split these developments in science, technology, and innovation. In this example, the science is in developing the math behind neural networks and understanding their workings. Technology is the implementation of neural networks and deep learning to pursue specific tasks in various fields. Innovation is the trial and error process of trying to integrate these new technologies within existing production processes and practices or the creation of new ones.

The current constellation between AI science, technology, and innovation is characterized by rapid progress of technologies. The mathematics of current AI breakthroughs was essentially ready by the late 1990s. But for these scientific insights to be translated into technology, material constraints in computing power and the size of available data sets need to be lifted. This happened through advances in computing, namely the availability and customization of graphics processing units for deep learning and the creation of massive sets of texts, speech, and image data that emerged as byproduct of widespread uses of digital devices and services (Halevy et al., 2009).

This also points to a challenge in the science of current AI. Cutting edge AI models are trained on massive amounts of predominantly proprietary data, demand for extremely high computing power, and abundant energy supply. There are shortages of the most advanced chips and computing systems, allowing the leading companies to demand a high premium. This puts scientists on their disinterested quest for knowledge and explanation at a disadvantage to corporate researchers, tasked with advancing technologies aimed to pursue specific functions and goals.

While early theoretical advances in neural networks stem from academic labs, their translation into technology and their use in production processes is very much concentrated within large companies (Metz, 2021). These companies have become dominant in the advancement of models (Ahmed et al., 2023). In practice, scientists need to collaborate with AI companies to access necessary compute and data to understand frontier systems and their workings.

This provides a challenge to science as the disinterested pursuit of understanding and explanation. If understanding and explanation depends on strong support of primarily commercially interested actors, who might face competitive or legal challenges due to scientific findings, disinterested pursuit of knowledge appears unlikely. Accordingly, knowledge about the workings, limitations, and effects of the most powerful AI systems will remain in private hands while public institutions will lose the ability to interrogate these systems, identify beneficial or detrimental effects, and regulate them appropriately. Consequently, if AI starts to figure more strongly in society, becomes more integrated into economics, politics, culture, and the military, governments will have to figure out how to provide scientists with access to models and compute, either through mandatory collaboration clauses for companies or through the financing and development of public compute clusters and models that are able compete with their commercial counterparts (Ho et al., 2021). Overall, this has developed into an area, where technology has leapfrogged science, in the sense that technology can do more than science can explain.

Breakthroughs in AI science, technology, and innovation are an important area for scientific enquiry. One obvious area for mathematics, computer science, and engineering is the understanding and explanation of model learnings and results. But there is also work for the social sciences. We have to identify the best conditions for achiev-

ing scientific breakthroughs in the field of AI. We also have to identify conditions under which new insights are translated into AI-enabled technology and where technology advances quicker than our understanding of it. Finally, we have to understand how novel insights and technologies are integrated into – and potentially transform – output functions for different actors in different fields. Analytically, these are questions about the workings, configurations, and constellations of structures, institutions, incentives, and policies within interconnected systems.

We need a science that helps explain the structures of successful environments for the generation of insights and the design of artifacts and processes in service of specific functions and goal (Narayanamurti & Tsao, 2021). We also need a better understanding of the impact of AI-technology when deployed in specific areas. Is the calculus between risks and opportunities the same when AI-enabled technology is deployed in different fields, such as the display of online ads or medicine? Clearly, this is not the case. We therefore need to understand the specific ways AI-enabled technologies are deployed in different fields, the functions they are supposed to fulfill, and the effects they actually have. This asks for less generalities and a greater focus on domain knowledge. This also translates into a greater emphasis on fact generation and documentation in innovation studies. We simply do not know very much about how people and organizations use AI-enabled technology in their output functions. Where are these technologies deployed and how does this – or does not – change outputs, processes, or practices. Without a more solid fact-base on how people and organizations integrate AI-enabled technology in the pursuit of their goals, we are flying blind about the actual and prospective impact of AI on society and risk mistaking innovation-speak for actual innovation (Vinsel & Russell, 2020).

Thinking through the subsequent impact of AI on different areas of society, we need to model technology and innovation specifically for each context, we are expecting AI to be used. The science of AI is the same across fields, but the determinants and dynamics of translating science into technology and innovation likely differ between different fields and need to be accounted for.

5 Adoption: Organizational and Individual

Novel breakthroughs in science, technology, and innovation alone do not guarantee widespread adoption or use (Edgerton, 1998/1999). In fact, most new technologies are never widely adopted, most innovations fail, and old technologies persist (Edgerton, 2007). Similarly, for AI adoption and widespread use are not a given. But prior waves of novel technologies and innovations provide templates of how to think about AI adoption by organizations and individuals.

Following Everett Rogers' influential work on the diffusion of innovation (Rogers, 1962/2003) extensive research has focused on organizations' adoption decisions of novel technologies and innovations (Tornatzky & Fleischer, 1990). This research explains organizations' decisions to adopt novel technologies and innovations with features on the demand as well as the supply side (Suriñach et al., 2009).

Demand-based explanations of adoption decisions focus on information availability, features of organizations making them more likely to attempt and achieve adoption, and constellations within the market environment an organizations finds itself in. Findings show that adoption decisions fundamentally depend on information being available

about novel technologies and innovations, their expected benefits, and ways to successful implementation (Bain, 1962; Bass, 1969; Griliches, 1957; Mansfield, 1961). Focusing on organizations, we see that early adoption profits from the type and quality of personell (Benhabib & Spiegel, 2005; Nelson & Phelps, 1966), amount of resources dedicated to research and development (Cohen & Levinthal, 1989, 1990), and the ability to adjust internal processes (Lane & Lubatkin, 1998; Lennox & King, 2004) and output functions to the opportunities and demands of novel technologies and innovations.

These factors point to a tension regarding the impact of organization-size on adoption. While large organizations might be better placed to adopt to novel technologies and innovations due to better information processing capabilities, dedicated personell, greater R&D resources, and standardized output functions, bureaucratic organization structures can struggle more with identifying, realizing, and adjusting to opportunities provided by novel technologies and innovations (Dougherty & Heller, 1994; Henderson & Clark, 1990). This inherent tension opens up the space for challengers to gain on incumbents due to earlier and more thorough adoption and adjustment to new opportunities provided by novel technologies and innovations. The market environment also matters for the adoption of innovation. Market share, relative competition, and prior adoption rate by market competitors all contribute to the likelihood of technology and innovation adoption (Gatignon & Robertson, 1989; Hannan & McDowell, 1984; Karshenas & Stoneman, 1993; Quirmbach, 1986; Reinganum, 1981).

Features in the supply of novel technologies and innovations also matter for adoption. This includes the degree to which suppliers can provide relevant information, continue development and provide updates, assist organizations in the adoption process (Rosenberg, 1972; Tonks, 1986). Accordingly, resources available to suppliers can positively impact adoption decisions.

This goes to show that the adoption of innovation is not a given but subject to many factors. To adopt novel technologies and innovations, organizations have to integrate new elements in their existing output functions or to establish new outputs from scratch (B. H. Hall & Khan, 2003). This carries uncertainty about associated gains and costs and makes adoption of novel technologies and innovations difficult, costly, and far from automatic. For one, most organizations have already processes, practices, and production factors set up that allow them to produce outputs and pursue goals. Any expected gains from AI must allow them to significantly outperform their existing processes to make up for costs of implementation, transition, and associated uncertainties. It is no surprise then to see large, well-resourced companies to experiment with novel technologies and practices early and thereby realizing associated competitive advantages over less well-resourced competitors. Making most novel innovations and technologies into a conservative force, allowing incumbents to extend their lead over the competition, as long as they maintain strategic awareness and openness (Porter, 1985/1998). Often, innovation is discussed as a potential disrupter of markets and power structures (Christensen, 1997/2024). But looking at the evidence above, disruptive innovation is the exception instead of the rule. Clearly, disruption is a possibility but subject to very specific contextual conditions.

Looking at individual decisions to use new technologies shows a similar picture. There are many models explaining technology adoption by people. But most are iterations of the theory of reasoned action by Martin Fishbein and Icek Ajzen (Fishbein & Ajzen, 1975, 2010). In its more recent iteration, the theory of planned behavior, this model holds that people are likely to act a given way if they belief this corresponds with the

preferences or behaviors of others and if they feel able to successfully perform the action as to achieve their intentions (Ajzen, 1991). Following this general model of behavior, we can explain AI adoption by individuals by their perception of how society and their most immediate environment thinks about AI use and someone's sense of efficacy in deploying AI successfully to achieve their goals.

These explanatory factors correspond well with the demand and supply logic, in the explanation of adoption decisions by organizations. Individual adoption decisions will be shaped by public discourse, prior adoption decisions and experiences among friends and family, and potentially deeper held attitudes toward advanced technologies. These are factors that allow people to assess the preferences of others toward their behavior. But demand for novel technologies and innovations is also driven by people's perceptions of whether they can use novel technology to achieve their goals. In a version of the Ajzen/Fishbein model focused on technological adoption, the technology acceptance model (TAM), this is captured by two variables: people's perception of a technologies usefulness in principle and the perceived ease of use (Davis, 1989; Davis et al., 1989). These are features of novel technologies and innovations that suppliers can shape. This includes user-conscious design, product iterations, and the availability of information resources.

These models, or some variations of them, have been successfully deployed in the analysis of adoption of various technologies. What they emphasize is the same basic picture that we have encountered with the adoption of technology by organizations: Innovation is not automatic and depends on perceptions of likely adopters, features of novel technologies and innovations, and contextual features. One crucial determinant for the adoption of novel technologies and innovations by organizations and people is their perception of what these enable them to do.

Technologies, be they artifacts or processes, have features that allow users specific actions (Gaver, 1991). One classic example comes from the design of everyday objects. A vertical door handle affords users to pull a door open, while horizontal plates afford pushing a door open (Norman, 1988/2014). These invited uses are called *affordances*. Originally, the term affordance stems from the psychology of perception (Gibson, 1979, 1967/1982). Following William H. Warren, (Warren, 1984, p. 683) we can define an *affordance* as the functional utility of an object that offers an actor certain action capabilities.

Affordances can be intentional or unintentional, and easy to perceive or hidden (Gaver, 1991). For example, a designer chooses a specific handle design to a teapot to have a user pour a cup in a specific way. This would be an intentional affordance. But there might be other features of the pot, which users discover through use and that allow them other actions not envisioned by the designer. For example, a high body might make it impractical for pouring tea but invite using the teapot as a flower vase. The easier an affordance is to perceive, the greater the ease of use, the greater the likelihood of adoption or compliance with the suggested action. This is crucial in the design of everyday objects or signs people need to perceive and understand quickly.

Complex technologies, like software systems, often carry many unintended affordances and can challenge users by hiding their intentional affordances. An example for an unintentional affordance of a software system is the user-driven development of mentions and hashtag-tagging on Twitter. These user-driven conventions provided the fledgling microblogging service Twitter with new affordances for interaction and discovery not initially thought-of by the service's designers.

For the adoption of novel AI-enabled technologies and innovations, we need to consider their affordances for likely adopters, be they organizations or individuals. The clearer would-be adopters perceive how they can use AI for their purposes, the easier they can implement AI in existing output functions or in the creation of new outputs. The greater their capability to find new affordances through experimentation, the more likely and widespread AI-adoption will be. Of course, the converse holds also. If AI-enabled affordances are hidden, difficult to realize, and experimentation remains elusive, the less likely is adoption.

This points to an inherent tension. The easier new opportunities are to perceive, the less transformative they are. And the most transformative ideas are, the harder they are to perceive – until they are not. For example, when electricity first was introduced people concentrated on electrify existing processes and machines. This resulted in some gains, but those were limited. But once new production functions were found, electricity became transformative (David, 1990). Similar patterns can be expected for AI. Right now many existing processes are in the process of being transformed into AI tasks. But once transformative new uses are found, we can expect much stronger gains and associated shifts (Agrawal et al., 2018/2022).

For AI-driven social change these patterns of technology and innovation adoption matter. Crucially, we cannot expect AI-enabled technologies to be widely adopted by all relevant actors nor that those who adopt them will be able to do so successfully. Instead, specific actors will adopt AI-enabled technologies and use them to optimize existing output functions or establish new ones. They will do so if they perceive AI to hold affordances that allow them to pursue their goals or provide them with a competitive advantage over others. Adopting AI will allow some of them to gain relative advantages over direct competitors or even improve their position of power within their action fields and networks. This happens either through purposeful strategic adaption to new affordances and contextual conditions introduced by novel technologies and innovations or they can passively profit from being already positioned in a way that allows them to capitalize on them without much adaption on their part.

6 Power Shifts

The study of power has many aspects and concerns; raising questions about its expression in coercion or voluntary cooperation, the legitimacy of authority and its challenge, or the pursuit and expression of power for its own sake (Arendt, 1969; Lukes, 2005; Poggi, 2001; Weber, 1921/2013). Of these approaches, the study of *power* as the “generalized capacity” (Parsons, 1963, p. 237) to achieve one’s goals successfully lends itself very well to answer the question of AI’s impact on power and power shifts within societies.

Michael Mann shows that people pursue goals through four distinct yet overlapping organizational networks: ideological, economic, military, and political (IEMP) (Mann, 1986/2012a, 1993/2012b, 2012c, 2013). At different times and under different conditions, these networks are more or less effective in enabling people to pursue their goals. They structure, rather than fully determine, how actors can exercise power by shaping their relative capacities to do so. This is not a static account, as the prominence and capability of these networks change over time. Mann attributes such shifts to uneven development across networks, driven by mechanisms such as organizational innovation, resource accumulation and conversion, and spatial expansion. These processes can alter

the balance among networks, shaping which actors and organizations rise or decline in relative prominence.

Building on Mann’s framework, we can think of AI as a contributing factor to organizational innovation, resource conversion, and spatial expansion that unevenly strengthens different actors and networks. By enhancing some actors’ organizational capacities more than others, AI can drive power shifts *within networks*. By accelerating uneven development across networks, it can also contribute to power shifts *between networks*. Shifts *within* networks reorder elites and organizations inside a given domain; shifts *between* networks reshape which domains are crucial to social coordination overall.

Even though the societal integration of AI is still in an early phase, observable patterns already align with these theoretical expectations. For instance, in military networks, AI-enabled systems reshape the relative capabilities of different weapon types leading to new investment patterns and tactics, illustrating power shifts within a network. Conversely, the emerging European debate on AI governance shows how technological change can alter the relative influence of political and economic networks shifting the power of civil rights and civil society actors to economic actors promising AI-driven growth, producing power shifts between them.

One area where AI has contributed to a shift *within* organizational networks is the military, where doctrine and procurement tend to follow shifting perceptions of contemporary and future warfare. In the 1980s, strategic missile defense initiatives such as the so-called “Star Wars” program and high-end platforms like the stealth bomber epitomized investments in large, capital-intensive systems. In the 1990s and early 2000s, debates about a “Revolution in Military Affairs” and network-centric warfare moved command, control and information flows to the centre of strategic thinking. The post-9/11 era saw an increased emphasis on counterterrorism, counterinsurgency, and special forces-driven operations. Today, AI-enabled capabilities are increasingly presented as the key to future warfare, whether in intelligence collection and analysis or in the coordination of automated and semi-autonomous units on the battlefield (Buchanan & Imbrie, 2022; King, 2025; Scharre, 2023). The current shift is driven not necessarily by extensive battlefield experience with AI but by expectations about future uses and capabilities. At the same time, recent experiences – such as the war in Ukraine, which has underscored the importance of drones and unmanned systems – reinforce hopes that AI will transform these domains (Horowitz et al., 2025; Onderco, 2025). Growing geopolitical tensions with China and concern about its industrial capacity have further fueled fears, especially in the United States, of a new arms race centered on AI (Milley & Schmidt, 2024; Neuberger, 2025). These developments, combined with advances in generative AI, have made AI a central reference point in thinking about future warfare and have strengthened the position of defense industries, doctrinal thinkers, and military branches associated with AI within what Mann would identify as the military organizational network.

We also find examples of how AI has contributed to power shifts *between* organizational networks in their relative prominence and influence. A prominent case concerns regulatory influence in the EU. In the early stages of the AI Act, civil society and rights-centered organizations – representatives of Mann’s ideological network – played a substantial role in shaping the regulatory agenda, risk categories, and rights-protective language (Palmiotto, 2025). As political actors increasingly framed AI in terms of economic competitiveness and national security, influence shifted toward the economic and military-political networks. Industry groups and security-oriented actors gained prominence in the later negotiations, while civil society actors saw their relative influence re-

duced, even if they continued to shape specific provisions and oversight debates (Csernaiti, 2025). Their subsequent pushback had limited effect. This broader shift in priorities is also reflected in emerging Commission discussions about aligning and potentially revisiting parts of the EU’s digital regulatory framework, including debates touching on the GDPR’s implications for AI development. Seen through Mann’s framework, changes in how AI is perceived as contributing to economic and military goals have elevated the economic and security networks in the formulation of collective rules – at the expense of actors that dominated the discourse only a few years earlier, when the risks to fundamental rights were more salient.

Leaving Mann’s original four networks but retaining his logic of uneven development, we can also observe a shift between organizational networks involved in the research and development of AI. Historically, AI research was largely driven by academic laboratories, where results, methods, and datasets were open in principle to scientific and public scrutiny. The resource intensiveness of contemporary frontier AI – massive compute requirements, large datasets, and the need for engineering at scale – has shifted much of the cutting-edge development to industry laboratories (Ahmed et al., 2023). As a consequence, companies now possess far greater capacity to set research agendas and determine which questions are pursued, while academic institutions retain strength mainly in foundational theory, safety research, and domain-specific applications. More precisely, while industry now leads in the development of large-scale frontier systems, academia remains central to foundational theory, methodological innovation, and domain-specific breakthroughs. In Mann’s terms, economic networks have accumulated and mobilized resources far more rapidly than academic-ideological networks, producing an uneven development in their respective capabilities to pursue research goals. This has generated a relative shift in power from university-based research to corporate AI labs, with implications for the types of knowledge produced and the degree of transparency with which it is shared.

Integrating this account of AI-driven power shifts into our broader model of AI-enabled social change clarifies both the mechanisms through which change may occur and the contingencies that shape when it does. The first contingency concerns the scale and direction of uneven development. AI can alter the relative position of actors within an organizational network – for example, by empowering some organizations more than others – but such within-network shifts do not necessarily transform broader social structures. By contrast, if AI produces significant shifts between organizational networks in their relative capacities to enable actors to pursue their goals, this can alter which networks become more central or peripheral to social coordination. For such shifts to translate into societal change, AI’s contribution must be substantial enough to outweigh countervailing factors, and the slope of uneven development must be markedly steeper in one network than in others.

More fundamentally, these shifts are not only a matter of technological potential but of political acceptance. Societies – and especially those actors who shape collective rules – may either accommodate or actively counteract emerging imbalances in organizational power. This raises the question of whether governance regimes are oriented toward preserving incumbent strengths or enabling challenger networks to rise. It is to this question of incumbent- versus challenger-friendly governance that we now turn.

7 Governance: Enabling challengers or supporting incumbents

The question of AI-enabled power shifts is not only a question of uneven adoption, fit, and adaptation between different actors and organizational networks; it is also a question of governance and regulation.

One important perspective on state governance highlights the relationship between governments and leading industries. Different political–economic systems structure these relationships in distinct ways (P. E. Hall & Soskice, 2001). In corporatist and coordinated market economies such as Germany, governments, employer associations, unions, banks and established industries are tightly interconnected. These dense networks support long-term coordination and institutional stability, but they also give incumbent firms significant influence over policy. As a result, protective interventions tend to favor existing actors, and adjustment often takes the form of incremental change within established structures rather than disruptive shifts (Streeck, 2009).

In contrast, liberal market economies, such as the United States, also feature close ties between government and industry but are more open to shifts within economic networks. They are more willing to embrace capitalist disruption. Governments continue to interact intensively with industry, but the type of industry they are interacting with changes over time as new sectors and firms gain prominence. The result is a greater openness to technology-enabled structural change (Phelps, 2006).

This can be seen, for example, when comparing the composition of the most R&D-intensive firms in the United States and Europe around 2003 and 2024. In the United States, the leading cohort has shifted markedly towards digital and platform companies; in Europe, legacy firms in car making and other traditional sectors have remained more prominent (Fuest et al., 2024). Put differently, the U.S. system is more accepting of incumbents being challenged and displaced, whereas European corporatist arrangements tend to offer greater protection to incumbents.

These differences in political–economic structure shape not only traditional industrial policy but also the governance of emerging technologies such as AI. If we translate this to the potential of AI-enabled social change, this potential clearly depends on systems and specific constellations of government–industry relationships, with some being more open to the emergence of challengers and others directly or indirectly supporting incumbents. AI-enabled shifts within and between organizational networks of power are mediated by states that vary in their willingness to accept or counteract these shifts. One arena, where this plays out is regulation and its impact on challenger/incumbent relationships and the constellations of actors and networks able to shape regulation (Fligstein & McAdam, 2012).

One way these broader political–economic constellations manifest themselves is through regulation. Regulation is often understood primarily as a means of protecting individual rights vis-à-vis the state or corporate actors – most visibly in areas such as privacy, consumer protection, or market regulation aimed at preventing monopolies and safeguarding competition. While this is an important dimension, regulation also plays a structural role in shaping the balance between incumbents and challengers. In many cases, regulatory frameworks tend to favor powerful incumbents – whether through market-access rules, compliance costs, or more diffuse mechanisms of regulatory capture (Carpenter & Moss, 2014). Although such arrangements can contribute to stability and continuity, they may also dampen competition, reduce incentives for innovation, and hin-

der the reallocation of resources toward actors with a better fit to emerging technological or economic demands. In this sense, regulation can become a mechanism that stabilizes existing power relationships and reinforces established industrial structures (Fligstein, 1996, 2001, 2005).

From this perspective, the expected societal impact of AI is likely to depend not only on technological capabilities but also on the specific regulatory choices made and the coalitions able to shape them. Recent tensions between frontier AI firms emphasizing self-imposed constraints and state actors seeking complete control and degrees of freedom in military integration and use further illustrate how governance mechanisms selectively favor actors whose organizational models align with dominant political–military priorities, even within ostensibly challenger-friendly systems (Hammond & Miller, 2026). Examining how regulation is designed, and by whom, can therefore offer insights into whether AI will act as a brake on change or as an enabler of structural transformation. Already at this early stage of broad societal and regulatory attention on AI, we can see these dynamics at play.

In recent years, there has been a clear shift in U.S. regulatory priorities around digital technology and AI. Under President Joe Biden, Democrats and many Republicans – albeit for different reasons – sought to curb the power of large technology companies, whether as alleged monopolists (through revitalized antitrust enforcement) or as perceived arbiters of speech (Kang, 2024). By contrast, the Trump administration has placed much greater emphasis on supporting U.S. leadership in frontier AI, treating leading technology firms as strategic national assets. Its AI executive orders and “America’s AI Action Plan” foreground deregulation, infrastructure and export promotion, and the framing of AI as a geopolitical race (Acton et al., 2025). Established technology companies have actively used narratives of geopolitical competition and national security to present themselves as indispensable to U.S. economic and military strength, and to argue for a permissive regulatory environment.

This shift has been visible, for example, in U.S. approvals for large exports of advanced AI chips and infrastructure partnerships with Gulf states, including Saudi Arabia, which directly benefit major U.S. semiconductor and cloud providers (Miller, 2025). At the same time, Washington and U.S. tech firms have pushed back against ambitious regulatory initiatives in Europe, especially the EU’s AI Act and related digital rules, contributing to ongoing debates in Brussels about delaying or softening parts of the regime (Smialek & Satariano, 2025). Taken together, these developments illustrate how regulatory and trade policies can be reoriented in ways that reinforce the position of already powerful incumbents in AI, once those firms succeed in framing their interests as aligned with national security and strategic competitiveness.

We see a related dynamic in the evolution of EU AI regulation. The EU AI Act was initially framed as an instrument to curb corporate power and safeguard fundamental rights, with an emphasis on transparency and accountability. Over time, however, successive revisions watered down some of the more far-reaching transparency and data-use constraints, especially for powerful general-purpose systems, and introduced broader space to justify data practices in the name of innovation and competitiveness. There may be good economic or innovation-oriented reasons for these adjustments, but the trajectory also illustrates how regulatory design responds to organized industry input (Moen & Heikkilä, 2025).

In the early phases of the AI Act, many large European industrial firms appeared only weakly engaged: AI regulation seemed to concern specialized developers and certain

high-risk sectors, rather than the core businesses of traditional incumbents. In that context, stricter limits on data-driven business models and AI deployment were politically easier to advance. As AI came to be seen as a core feature of future products, processes, and competitiveness, industrial incumbents began to align more explicitly with European AI developers and to intervene more forcefully in the regulatory process. Regulatory priorities shifted accordingly: from a primarily rights- and consumer-protection-oriented framing toward a stronger emphasis on industrial policy, competitiveness, and “digital sovereignty” as overarching goals. In other words, as the cost–benefit calculation of major incumbents changed, the balance of the EU’s regulatory agenda changed with it.

Taken together, the US and EU cases suggest that powerful industry incumbents on both sides of the Atlantic are actively building AI capabilities while simultaneously using their relationships with governments to shape the rules under which AI develops. AI may transform how these firms operate and generate profits, but by embedding themselves early and deeply in regulatory processes, they attempt to preempt more disruptive shifts in market structure that would empower new challengers at their expense.

Within the broader model of AI-enabled social change, these examples highlight two complementary points. First, governance and regulation are themselves shaped by AI-driven shifts in organizational fields and by the strategies of incumbents and challengers. Second, existing governance arrangements and regulatory choices condition the trajectory of technological development and diffusion: they structure who can experiment, which business models are viable, how individuals adopt or resist AI systems, and where uneven developments and new forms of power concentration emerge.

8 Constraints

The preceding discussion emphasized perceptual, organizational, and regulatory mechanisms that shape AI-enabled social change. These mechanisms operate within a broader set of structural constraints that condition societies’ ability to develop, deploy, and benefit from AI. These constraints fall into three interconnected categories – material, institutional, and societal – that together shape both the opportunities and the limits of AI adoption across different political and economic systems.

Material constraints consist of tangible infrastructures and resources that make AI possible. Although public debate often foregrounds algorithms, the societal use of AI is fundamentally dependent material factors, such as access to advanced models, computational resources, energy, and human talent (Ho et al., 2021). Only a small number of technologically advanced states are likely to develop frontier models themselves, while most will depend on foreign-developed systems as a basis for localized applications. This dependence is increasingly viable, given the expanding availability of models, but it introduces structural dependencies on foreign companies and states, raising concerns about security, continuity of access, and long-term technological sovereignty. Similar constraints apply to the computational infrastructure required for AI development. High-performance chips optimized for training and inference remain scarce and are produced in highly concentrated global supply chains that are vulnerable to geopolitical tensions and export controls (Miller, 2022). Even where financing is available, access to such hardware is uneven and subject to disruptions. Moreover, data centers rely on substantial and stable sources of energy. Meeting these needs often requires expanding national grids or establishing dedicated power generation, which increases costs

and may conflict with environmental, climate, or local land-use priorities. In many countries, including those pursuing ambitious decarbonization strategies, energy-intensive AI infrastructure can clash with established policy goals, creating political friction that slows or limits implementation. Finally, societies depend on the availability of human talent to develop, run, and maintain different layers of the AI stack. These material constraints therefore form a foundational layer that shapes which societies can develop meaningful AI capabilities and at what scale.

A second layer of constraints is institutional, operating through the regulatory and governance frameworks that define permissible forms of AI development and deployment. Regulation can directly limit the availability of training data, set boundaries on what constitutes acceptable AI behavior, and determine which AI-enabled services can be offered. Such rules can be enabling, but they may also restrict experimentation or impose significant compliance burdens. In the European context, for instance, data protection regimes have historically limited the availability and use of data crucial for AI training (Jia et al., 2021, 2025), especially before more flexible interpretations introduced through reforms such as the 2025 Digital Omnibus. Institutional constraints, however, also operate indirectly. Regulatory ambiguity – uncertainty about legal obligations, unclear enforcement practices, or evolving rules – can create perceived risks that discourage firms, public organizations, and investors from experimenting with new technologies. Early-stage innovation is particularly sensitive to such uncertainty, and a regulatory environment seen as unpredictable may redirect investment or slow organizational learning. Because early experimentation enables societies to understand how AI can be adapted to specific contexts, institutional constraints can shape not only the pace of AI diffusion but also the development of robust, domain-specific forms of AI implementation.

Finally, societal constraints concern the broader political, cultural, and economic environments within which AI adoption unfolds. The availability of skilled labor across the AI stack – from model development to domain-specific integration – depends on educational systems, labor market structures, and societal expectations regarding the value and future prospects of AI-related competencies. Without sufficient human capital, even well-resourced societies may struggle to realize AI’s potential. Public narratives and political discourse also play a crucial role. Whether AI is framed primarily as an opportunity or as a threat influences investment decisions, organizational experimentation, and political support for AI-related infrastructure. Societies that emphasize risks may adopt more cautious stances, slowing learning processes and reducing uptake, while more optimistic environments may foster early adoption and rapid diffusion. Political systems themselves introduce additional variation: centralized systems may mobilize resources more quickly, whereas pluralistic democracies may move more slowly but benefit from greater openness and diversity of experimentation. Recent political shifts in some democracies – toward polarization, regulatory uncertainty, or fragmented governance – may weaken innovation ecosystems that historically supported technological leadership.

Taken together, these material, institutional, and societal factors constitute a complex constellation of constraints that shape how, where, and at what pace AI can be developed and integrated. They influence not only the availability of AI capabilities but also the ways societies interpret, adapt, and regulate them. Understanding AI-enabled social change therefore requires attention to these structural conditions as integral components of the broader causal framework. Future research should examine how societies attempt to navigate, mitigate, or restructure these constraints – and how such efforts interact with political, economic, and technological developments over time.

9 Conclusion

This article has advanced a theory of AI-induced social change that provides a structured roadmap for both research and public debate. Current discussions about AI largely replay familiar patterns from earlier waves of digital transformation. Public and scholarly discourse oscillates between extremes (Jungherr et al., 2020): on the one hand, sweeping claims that AI will fundamentally and comprehensively transform society; on the other, dismissals of AI as little more than hype, a marketing narrative driven by technology firms whose only tangible effects are marginal efficiency gains and new forms of inequality. While these polarized narratives are effective at generating attention and controversy, they hide the actual mechanisms through which technologies shape societies and power relations. They also hinder systematic empirical analysis by conflating localized change with global transformation.

The theory developed here rejects both technological determinism and technological trivialization. Instead, it conceptualizes AI-driven social change as contingent, stepwise, and mediated. AI does not automatically transform societies simply through its technical capabilities. Nor does its impact stop at superficial adoption or symbolic use. Rather, AI alters social outcomes through a chain of interlinked processes that connect scientific breakthroughs, technological innovation, adoption by organizations and individuals, shifts in relative power within and between organizational networks, governance responses, and structural constraints. At each step, the continuation of the chain is conditional. Breaks, bottlenecks, or countervailing forces can halt or redirect potential trajectories of change.

At its core, the theory conceptualizes AI-induced social change as a multi-level stepwise causal process. Innovations in AI science and technology create new affordances, but these affordances only matter insofar as actors perceive them, adopt them, and integrate them into their output functions. Adoption is uneven, favoring actors whose organizational structures, resources, and institutional environments align well with AI-enabled affordances. This uneven adoption can generate relative advantages, which may translate into power shifts within organizational networks (such as markets, bureaucracies, or military institutions) or between different sources of social power.

Crucially, these shifts are neither automatic nor self-reinforcing. They trigger governance responses, as actors who benefit from AI seek permissive regulatory environments while those disadvantaged by AI demand restrictions or compensatory interventions. Governance, in turn, reshapes the conditions under which future AI development and adoption occur. Throughout this process, material, institutional, and societal constraints shape what is possible. The result is a recursive, path-dependent process rather than linear or inevitable transformation.

The dynamics emphasized by this framework help explain why AI can simultaneously appear transformative in some contexts and inconsequential in others. AI may substantially alter localized production functions, organizational practices, or competitive dynamics without producing immediate or visible macro-level change. For example, AI-enabled production functions can allow specific firms, political organizations, or military units to outperform competitors. This constitutes a real and empirically observable change at the micro or meso level. Yet such changes do not necessarily aggregate into systemic social transformation. They may be contained by regulation, offset by countervailing adaptations, or limited to specific domains.

Conversely, the absence of dramatic macro-level change should not be interpreted as

evidence that AI has had no impact. Localized shifts in practices, incentives, and power relations can accumulate over time or remain consequential within bounded fields. The framework thus asks for analytical precision: change at one level does not imply change at all levels, and stability at the macro level does not negate meaningful transformation at lower levels.

Like any theory, the framework proposed here can fail: It would fail if AI were to produce rapid, comprehensive social transformation without passing through the intermediate steps of uneven adoption, power shifts, and governance mediation outlined in the model. It would also fail if AI turned out to have no durable impact on production functions, organizational capacities, or relative power relations beyond short-lived experimentation and symbolic use. Finally, the theory would be challenged if empirical research consistently showed that AI adoption and effects were uniform across actors and contexts, leaving no room for the differentiated and conditional dynamics emphasized here.

However, current evidence points in the opposite direction. Early patterns of AI development and use already show strong unevenness across sectors, organizations, and societies, as well as intense political contestation over governance and regulation. These observations suggest that the theory captures central features of AI's societal trajectory, even if specific pathways and outcomes remain uncertain.

The theory advanced in this article is not intended as a final word on AI and social change. Rather, it offers a conceptual framework for cumulative research. Its primary contribution lies in linking specific phenomena across analytical levels (i.e. micro, meso, and macro) and clarifying the conditions under which localized changes can, or cannot, translate into broader social transformation.

For empirical research, this framework provides a way to situate focused studies within a larger causal chain. Careful empirical work necessarily concentrates on specific mechanisms, contexts, or outcomes. Such focus is a strength, not a weakness. But without a larger theory, individual findings risk being overgeneralized or misinterpreted. By mapping where a given phenomenon sits within the broader process of AI-induced change, researchers can better assess its significance, limits, and downstream implications.

More broadly, the theory underscores that the societal impact of digital technologies, and AI in particular, is differentiated and conditional. Understanding this differentiation is not merely an academic exercise; it is essential for informed governance and public debate. Only by identifying where change occurs, for whom, and under what conditions can societies meaningfully assess AI's consequences and shape its development in line with collective goals.

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